

Lab 9 Individual Contribution Report

Application and Deployment Enhancement

CS 5542 — Big Data and Analytics | Spring 2026

Name	Tony Nguyen
GitHub Handle	mosomo82
Team	Tony Nguyen · Daniel Evans · Joel Vinas
Project	HyperLogistics / Multi-Agent Trucking Analytics Platform

1. MY ROLE ACROSS LABS 6–8

In Lab 6 I served as AI Infrastructure lead — designed the Gemini 2.5 Flash agent, built the first five analytical tools, and deployed the Streamlit dashboard to Streamlit Cloud. In Lab 7 I led reproducibility infrastructure (Dockerfile, Makefile, 17-test audit suite). In Lab 8 I owned the PEFT fine-tuning pipeline: scripted the dataset generation from Snowflake SILVER tables, tested QLoRA configurations, and managed adapter training on Colab. For Lab 9, I am the lead on all cross-lab documentation and the Lab 8 critical fixes.

2. PERSONAL CONTRIBUTIONS — LAB 9

2.1 Lab 8 Critical Fixes (My Primary Lab 9 Focus)

Lab 8 was my primary contribution in Lab 9. I identified and fixed all three critical issues from the Lab 8 evaluation:

- **CPP Spatial SQL hard gate:** Implemented the deterministic bridge constraint check in `sf_connect.py` before any LLM invocation. The Spatial SQL query retrieves `MIN(weight_limit_tons)` and `MIN(height_limit_ft)` across all bridges on the candidate route. If the vehicle exceeds either limit, a **HARD VETO** is returned without calling Phi-2. This fully resolves the Q13→Q14 monotonicity failure on both Colab and RTX 3060.
- **Evidence injection:** Updated the prompt template in `prompt_adaptation.py` to include a `[RETRIEVED CONSTRAINTS]` block populated by the Step 3a SQL query. The model is explicitly instructed not to recall physical limits from memory, only from the injected block. Hallucination rate on the expanded 50-query evaluation drops from ~40% to 8%.
- **Accuracy discrepancy:** Corrected the Section 5 intro in the group report (60% vs 40% baseline) and added an explanation of the Colab vs RTX 3060 PEFT performance gap to `EVALUATION.md`.
- **CoT-formatted training data:** Updated `generate_dataset.py` to produce full 4-step reasoning chains (Disruption → Route → Constraint → Decision) in all output fields. Expanded training dataset from 100 to 300+ pairs covering 5 disruption categories. PEFT CoT quality improves from 67% to 83% with the new training data.
- **Evaluation expansion:** Rewrote `evaluation.py` to implement the 5-dimension rubric scorer (0–10) and expanded the evaluation set from 15 to 50 queries across 5 disruption types. Metamorphic test suite expanded from 3 to 13 pairs.
- **CI integration:** Authored `.github/workflows/lab8_eval.yml` for mock-mode evaluation on every push, consistent with the CI patterns established in Labs 6 and 7.

2.2 Lab 8 Documentation

- ARCHITECTURE.md (Week_8/) — Complete 4-layer CPP architecture reference: data perception layer (SILVER tables), intelligence layer (ReMindRAG + SRSNet), validation layer (CPP Step 3a + Step 3b), application layer (Streamlit). Includes failure mode analysis and mitigation table for all three critical Lab 8 issues.
- EVALUATION.md (Week_8/) — Corrected results table, rubric scoring methodology, expanded 50-query suite design, CI workflow YAML, and improvement traceability table.

2.3 Lab 6 Documentation and Infrastructure (from previous section)

- ARCHITECTURE.md and EVALUATION.md (Week_9/) responding to Lab 6 instructor feedback on architecture and evaluation documentation.
- Snowflake keep-alive ping, retry decorator, FastAPI /health endpoint, dependency pinning, structured logging, tool-call trace expander, sidebar UI, expanded eval suite (5 → 10 scenarios) with CI.

2.4 Lab 7 Benchmarking and Verification (from previous section)

- BENCHMARKING.md — Constrained LooGLE (F1 = 0.496) and HotpotQA (F1 = 0.58) runs; efficiency verification; variance table; JSON output schema.
- REPRO_VERIFICATION.md — 9 new automated tests (test_benchmark_smoke.py + test_repro_variance.py); suite 17 → 26 tests; CI integration at zero API cost.

3. GITHUB CONTRIBUTION EVIDENCE

File / Artifact	Description	Repo
Week_8/ARCHITECTURE.md	4-layer CPP architecture + failure mode analysis (new)	COMP_SCI_5542
Week_8/EVALUATION.md	Corrected results + rubric + 50-query expansion (new)	COMP_SCI_5542
Week_8/app/evaluation.py	5-dim rubric scorer; 50-query set; 13 metamorphic tests	COMP_SCI_5542
Week_8/data/generate_dataset.py	CoT-formatted outputs; 300+ pairs; 5 disruption types	COMP_SCI_5542
Week_8/adaption_method/prompt_adaptation.py	[REMOVED CONSTRAINTS] evidence injection (new)	COMP_SCI_5542
Week_8/scripts/sf_connect.py	CPP Spatial SQL hard gate (Step 3a)	COMP_SCI_5542
.github/workflows/lab8_eval.yml	GitHub Actions CI: mock-mode eval + 70% parallelism	COMP_SCI_5542
Week_9/ARCHITECTURE.md	Lab 6 architecture reference (Lab 6 feedback)	COMP_SCI_5542
Week_9/EVALUATION.md	Lab 6 eval methodology (Lab 6 feedback)	COMP_SCI_5542
BENCHMARKING.md	Paper-vs-repro metrics (Lab 7 feedback)	ReMindRAG_Week7
REPRO_VERIFICATION.md	26-test suite expansion (Lab 7 feedback)	ReMindRAG_Week7
tests/test_benchmark_smoke.py	5 pipeline output quality tests	ReMindRAG_Week7
tests/test_repro_variance.py	4 cross-run determinism tests	ReMindRAG_Week7
.github/workflows/ci.yml	Unified CI: Lab 6 eval + Lab 7 smoke + variance	Both

Main Repo: https://github.com/mosomo82/COMP_SCI_5542

ReMindRAG Repo: https://github.com/mosomo82/ReMindRAG_Week7

Commit History: https://github.com/mosomo82/COMP_SCI_5542/commits?author=mosomo82

4. TOOLS USED

Tool / Platform	Purpose
Claude (Anthropic)	Report drafting; all six architecture/evaluation doc files; CI workflow authoring
Cursor	AI-assisted code editing for CPP Spatial SQL gate, evidence injection, rubric scorer
GitHub Actions	CI: Lab 6 eval (10 scen.) + Lab 7 smoke/variance + Lab 8 mock-mode eval
Google Colab (T4)	QLoRA fine-tuning of Phi-2; original PEFT adapter training
Streamlit Cloud	Deployment for both Lab 6 and Lab 8 live apps
Snowflake	Data warehouse for all logistics tables (both labs)
Gemini 2.5 Flash	AI agent for Lab 6 trucking analytics platform
Phi-2 + PEFT/QLoRA	Domain-adapted safety validation agent (Lab 8)
bitsandbytes / TRL	4-bit NF4 quantization and QLoRA training infrastructure
pytest	All 26 automated reproducibility + benchmark tests (Lab 7)
FastAPI	Lab 6 /health sidecar endpoint
Docker	Dependency pinning validation (both labs)

5. SUMMARY

For Lab 9 I contributed across all three prior labs. My primary focus was Lab 8: I implemented the CPP Spatial SQL hard gate that resolves the monotonicity failure, the evidence injection pattern that drops hallucination rate from 40% to 8%, the CoT-formatted dataset expansion (100 → 300+ pairs), and the rubric-based evaluation suite (15 → 50 queries). I also authored the two Lab 8 documentation artifacts (ARCHITECTURE.md, EVALUATION.md), the two Lab 6 artifacts (in response to instructor feedback), and the two Lab 7 artifacts (BENCHMARKING.md, REPRO_VERIFICATION.md). All CI workflows across the three labs were authored and integrated by me, giving the team a unified automated quality gate for the Research-A-Thon.